Technical Paper: Payment Transaction Anomaly Classification

Abstract

This paper presents an end-to-end machine learning solution for classifying payment transactions as legitimate or fraudulent. Using a real-world, highly imbalanced dataset of 284,807 European credit card transactions, a Logistic Regression model with balanced class weighting achieved a Recall of 87.4% and ROC-AUC of 0.966, demonstrating that a foundational, interpretable model can effectively flag likely fraud despite extreme class imbalance.

Introduction

Payment fraud detection is a critical, widely studied problem in financial technology. Fraudulent transactions are rare relative to legitimate ones, making this a naturally imbalanced classification problem. This project investigates whether a foundational, zero-cost ML approach can meaningfully separate fraudulent from legitimate transactions using only anonymized transaction-level features.

Problem Definition

Given a transaction's amount, timing, and a set of anonymized behavioral features, predict whether the transaction is fraudulent (Class = 1) or legitimate (Class = 0).

Related Work

Fraud detection is commonly approached with foundational models such as Logistic Regression, Decision Trees, and ensemble methods, often combined with resampling techniques (e.g. SMOTE, undersampling) or cost-sensitive learning to address class imbalance. This project uses class-weighting as a simple, effective baseline technique for the same purpose.

Dataset

The Credit Card Fraud Detection dataset (ULB Machine Learning Group, mirrored by TensorFlow) contains 284,807 transactions with 31 columns: `Time`, `Amount`, 28 PCA-transformed features (`V1`–`V28`), and the binary target `Class`. Only 492 transactions (0.17%) are fraudulent.

Methodology

1. **Data Cleaning:** Checked for missing values (none found) and duplicates (1,081 found and removed).
2. **Preprocessing:** Scaled `Amount` and `Time` independently using StandardScaler; PCA features were used as provided.
3. **Splitting:** 80/20 train-test split, stratified by class to preserve the fraud ratio in both sets.
4. **Modeling:** Logistic Regression with `class_weight='balanced'` to compensate for the severe class imbalance without discarding data.

EDA

Exploratory analysis confirmed the extreme class imbalance (0.17% fraud) and showed that fraudulent transactions have a visibly different distribution of transaction amounts compared to legitimate ones, motivating the inclusion of `Amount` as a scaled feature.

Model Development

A single foundational model (Logistic Regression) was selected for its interpretability and suitability given the assignment's scope, with class weighting as the primary imbalance-handling technique.

Results

Metric	Value
Accuracy	97.52%
Precision	5.61%
Recall	87.37%
F1-score	10.55%
ROC-AUC	0.966

Confusion Matrix:

```
[[55255 1396]
[ 12 83]]
```

Discussion

Accuracy alone (97.5%) would be misleading here, since a model that always predicts "legitimate" would already achieve ~99.8% accuracy while catching zero fraud. The meaningful metrics are Recall (87.4%) — the model catches the large majority of actual fraud — and ROC-AUC (0.966), indicating strong overall discriminative power. Precision is low (5.6%), meaning many flagged transactions are false alarms; this Recall/Precision trade-off is a deliberate and common choice in fraud detection, where missing fraud is typically far costlier than investigating a false positive.

Limitations

• Low Precision would require further tuning (e.g. threshold adjustment, resampling) before practical deployment.

• The anonymized PCA features (V1–V28) limit interpretability of individual feature contributions and restrict the manual-input demo app to only two truly adjustable fields (Amount, Time).

• This is an educational prototype and has not been tested against adversarial or evolving fraud patterns.

Future Scope

• Compare against Decision Tree or ensemble methods (e.g. Random Forest).

• Apply SMOTE or other resampling strategies to improve Precision.

• Explore cost-sensitive threshold tuning based on business impact.

Conclusion

A foundational Logistic Regression model, combined with careful class-imbalance handling, can achieve strong fraud-detection performance

(87% Recall, 0.966 ROC-AUC) on a real-world imbalanced dataset — showing that effective fraud detection does not necessarily require complex deep-learning approaches for a first working solution.

References

• Dataset: Credit Card Fraud Detection Dataset, ULB Machine Learning Group, mirrored by TensorFlow: https://storage.googleapis.com/download.tensorflow.org/data/creditcard.csv

• Scikit-learn documentation: https://scikit-learn.org/